



Inspiring Excellence

**CSE370 : Database Systems
Project Report**

**Project Title : Lost & Found Pet Tracking & Recovery
Platform**

Group No : 5, CSE370 Lab Section : 17, Spring 2026		
ID	Name	Contribution
23301323	Rafid Rohan	Average
23301569	Md. Rahatul Islam Rahat	Average
23301325	Sk. Md. Rashid Assef Shibly	Average

Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Project Features	4
3	ER/EER Diagram	5
4	Schema Diagram	6
5	Normalization	7
6	Frontend Development	8
7	Backend Development	12
8	Source Code Repository	14
9	Conclusion	14
10	References	14

Introduction

The Lost & Found Pet Tracking & Recovery Platform is a database-driven system designed to manage lost pet reports, found pet sightings, and recovery processes in a centralized and structured manner.

Currently, pet owners and finders rely on scattered platforms such as social media, which makes tracking and matching cases inefficient and time-consuming. This system addresses the problem by storing all reports in a relational database and enabling automated matching between lost and found cases.

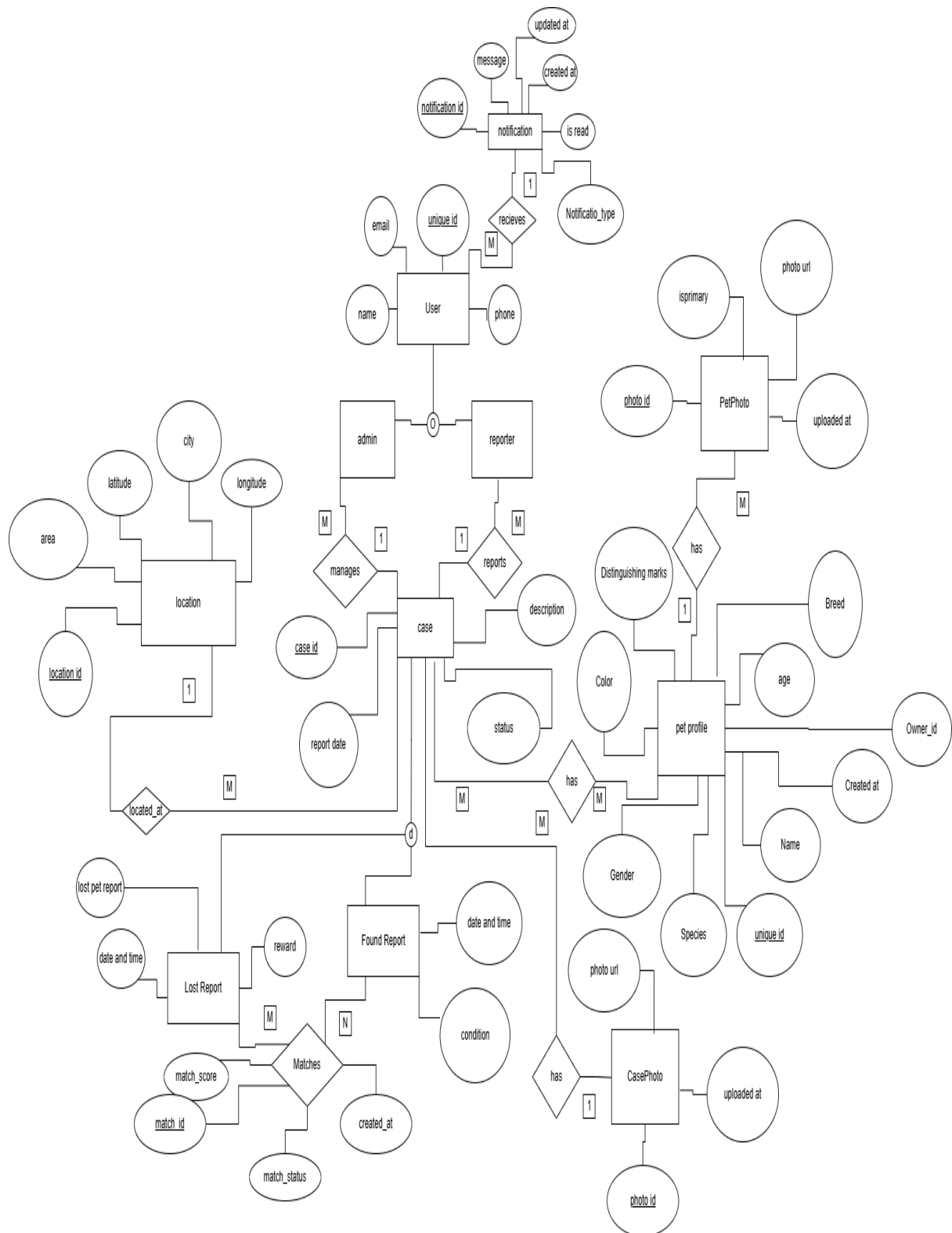
The platform supports structured case registration, pet profile management, automated match suggestion, and case lifecycle tracking. Additionally, it integrates location-based querying and relational linking between cases to improve recovery efficiency.

The goal of this system is to reduce manual searching effort, improve match accuracy, and provide a reliable and organized solution for pet recovery.

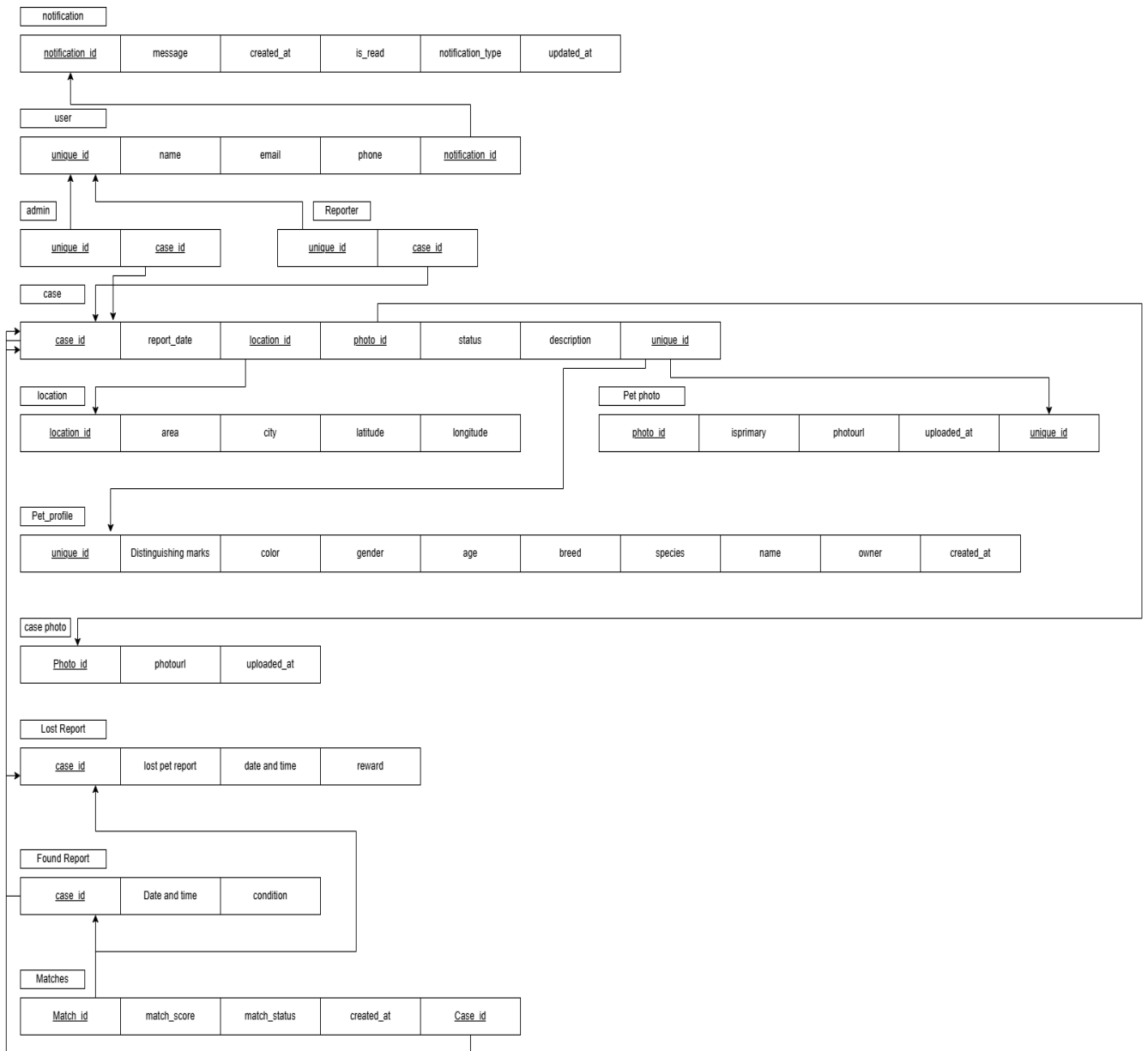
Project Features

ID, Name	Features [3 per member]	
23301323, Rafid Rohan	Ft 1	Dashboard & System Overview Interface
	Ft 2	Analytics & Reporting Module
	Ft 3	Pet Profile Management System
23301569, Md. Rahatul Islam Rahat	Ft 1	Lost Pet Case Registration System
	Ft 2	Found Pet Case Registration System
	Ft 3	Case Management System (Update & Deletion)
23301325, Sk. Md. Rashid Assef Shibly	Ft 1	Automated Match Suggestion System
	Ft 2	Notification Management System
	Ft 3	Active Case Monitoring System

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Yes it was in 1NF, cause in petprofile table we could have multiple distinguishing marks or color. For that reason we made another table for them. Yes, It was in 2NF. but violets the law of 3NF, because we must get city info first before getting area in location table. there comes a transitive dependency. Thus we made another table for city info.

Frontend Development

The frontend of the Lost & Found Pet Tracking & Recovery Platform was developed to provide a clean, user-friendly, and responsive interface for users to interact with the system efficiently.

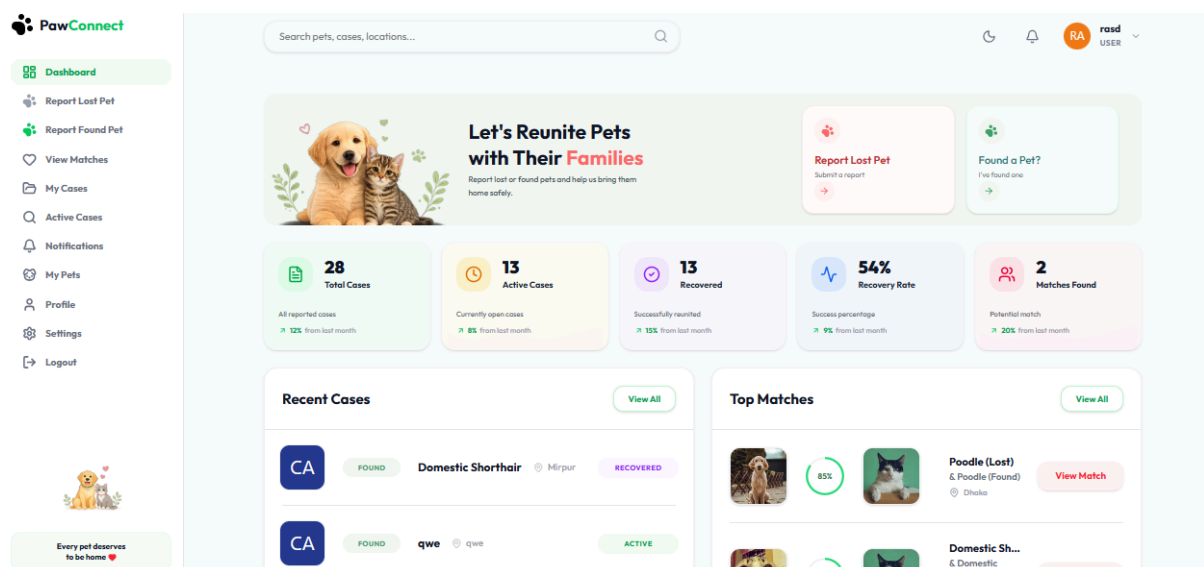
The interface includes key modules such as dashboard overview, lost and found reporting forms, pet profile management, case tracking, and match viewing. The design focuses on simplicity and usability, allowing users to quickly report cases, browse matches, and track recovery progress.

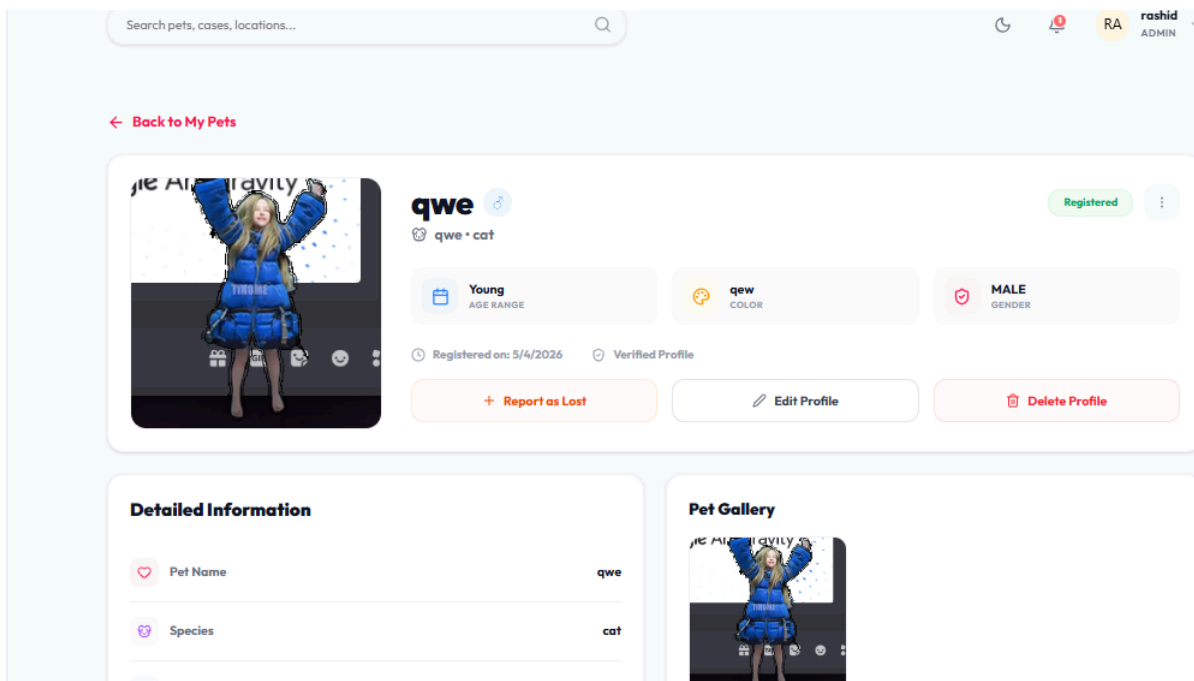
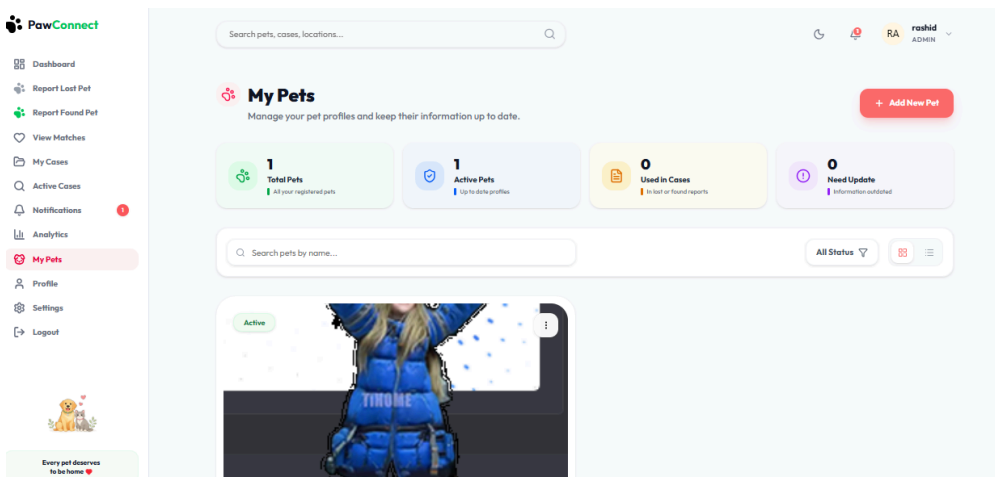
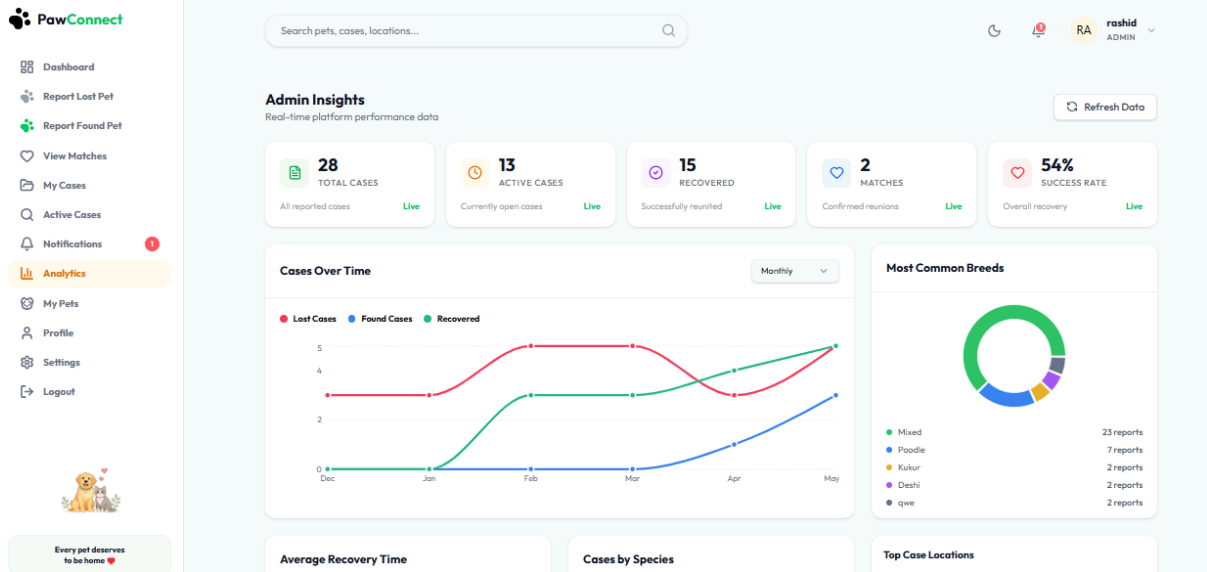
Modern UI practices were followed to ensure consistency across pages, including structured layouts, intuitive navigation through a sidebar, and clear visual hierarchy. The system supports both light and dark modes to enhance user experience.

Frontend components dynamically fetch and display data such as active cases, notifications, and analytics, ensuring that users always see updated information.

Contribution of ID: 23301323, Name: Rafid Rohan

- Designed and implemented the main dashboard interface
- Developed analytics visualization UI
- Built pet profile management pages (view, add, edit)





Contribution of ID: 23301569, Name: Md. Rahatul Islam Rahat

- Implemented lost and found reporting forms
- Designed case management interface (update, delete)
- Worked on input validation and form usability

The screenshot shows the 'Report Found Pet' form. At the top, there is a search bar and a user profile for 'rashid' (ADMIN). The form is titled 'Report Found Pet' with a subtext 'Thank you for helping a pet find its way back to its family.' The form is divided into two main sections: 'Pet Information' and 'Photo Evidence'. The 'Pet Information' section includes fields for 'Pet Name' (placeholder: 'Unknown (or temporary name)'), 'Species' (dropdown: 'Select Species'), 'Breed / Description' (placeholder: 'e.g. Golden Retriever or Mixed Breed'), 'Gender' (dropdown: 'Unknown'), 'Age Range' (dropdown: 'Adult (3-8 years)'), 'Primary Color(s)' (placeholder: 'e.g. Golden / White with Brown patches'), and 'Distinctive Marks / Features' (placeholder: 'e.g. White spot on chest, wearing a red collar...'). The 'Photo Evidence' section has a heading 'Photo Evidence' with the subtext 'MULTIPLE PHOTOS ENCOURAGED'. It features a 'Click to Upload Photos' button with a camera icon and the text 'MULTI-PHOTO SUPPORT' and 'Select up to 10 photos of the pet'. Below this, there is a tip: 'High quality photos increase recovery chances by up to 80%.' and a list of tips: 'Clear facial features', 'Visible unique marks', and 'Good natural lighting'. At the bottom of the form, there is a 'Submit Report' button and a 'Reset Form' button. A disclaimer at the bottom states: 'By submitting, you agree to our terms. We'll start matching your report immediately.'

The screenshot shows the 'Report Lost Pet' form. At the top, there is a search bar and a user profile for 'rashid' (ADMIN). The form is titled 'Report Lost Pet' with a subtext 'Let's help you bring your furry friend back home safely.' The form is divided into two main sections: 'Pet Information' and 'Photo Evidence'. The 'Pet Information' section includes fields for 'Pet Name' (placeholder: 'e.g. Bruno'), 'Species' (dropdown: 'Select Species'), 'Breed / Description' (placeholder: 'e.g. Golden Retriever or Mixed Breed'), 'Gender' (dropdown: 'Unknown'), 'Age Range' (dropdown: 'Adult (3-8 years)'), 'Primary Color(s)' (placeholder: 'e.g. Golden / White with Brown patches'), and 'Distinctive Marks / Features' (placeholder: 'e.g. White spot on chest, wearing a red collar...'). The 'Photo Evidence' section has a heading 'Photo Evidence' with the subtext 'MULTIPLE PHOTOS ENCOURAGED'. It features a 'Click to Upload Photos' button with a camera icon and the text 'MULTI-PHOTO SUPPORT' and 'Select up to 10 photos of the pet'. Below this, there is a tip: 'High quality photos increase recovery chances by up to 80%.' and a list of tips: 'Clear facial features', 'Visible unique marks', and 'Good natural lighting'. At the bottom of the form, there is a 'Submit Report' button and a 'Reset Form' button. A disclaimer at the bottom states: 'By submitting, you agree to our terms. We'll start matching your report immediately.'

Search pets, cases, locations...

rashid

ADMIN

System Case Management

Administrator view: Managing all pet recovery cases across the platform

ADMIN LIVE MONITOR

All Cases 28

Lost 24

Found 4

Sort by: Newest

Search by breed or city...

Pet Info	Case Details	Date	Status	Actions
PET Domestic Shorthair Male • Baby • Orange	Dhaka 4000048	May 04, 2026 01:33 AM	Recovered	View Details
PET qwe Male • Baby • qwer	qwe 4000039	May 04, 2026 01:31 AM	Reported	View Details
PET re Female • Baby • qwer	rqwe 4000038	May 04, 2026 01:24 AM	Reported	View Details
PET asr Female • Baby • awar	ras 4000037	May 04, 2026 01:24 AM	Reported	View Details
Domestic Shorthair Male • Young • Orange \$500.00	Dhaka 4000036	May 04, 2026 12:33 AM	Recovered	View Details
Mixed Male • 2 • Brown \$500.00	Badda 4000035	May 03, 2026 06:50 AM	Recovered	View Details

Contribution of ID: 23301325, Name: Sk. Md. Rashid Assef Shibly

- Developed match viewing interface
- Implemented notification UI system
- Built active case tracking and status display components

Search pets, cases, locations...

rashid

ADMIN

Admin Management Mode

Viewing and managing all potential matches across the platform.

PLATFORM VIEW

All Matches 2

High Confidence

Medium Confidence

Low Confidence

85% Match

CONFIRMED

LOST

85% SCORE

FOUND

Male • Adult

Dhaka

Male • Adult

Dhaka

Matched 6/27/2026

View Details

View Contact Details

75% Match

CONFIRMED

LOST

75% SCORE

FOUND

Male • Young

Dhaka

Male • Baby

Dhaka

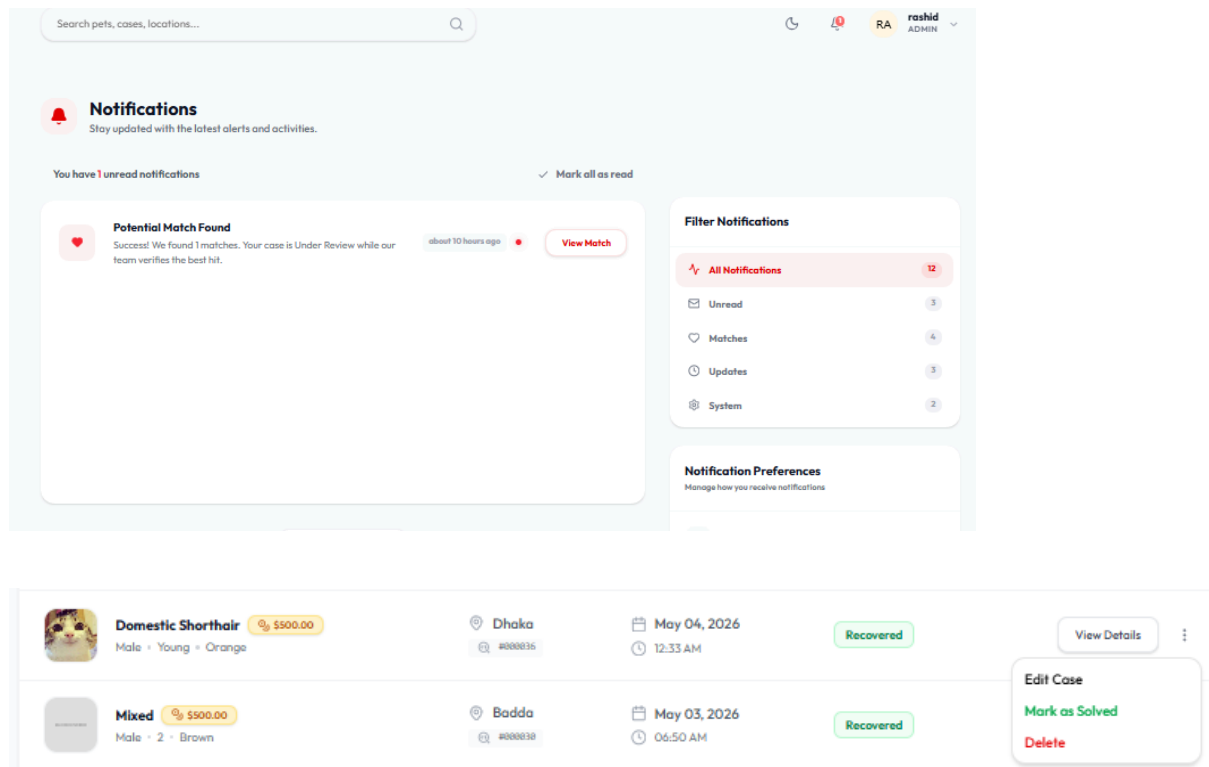
Matched 5/4/2026

View Details

View Contact Details

Tip

Review the details carefully before confirming a match. You can contact the other party securely through the platform using the Review Match button.



Backend Development

The backend of the Lost & Found Pet Tracking & Recovery Platform was designed using a relational database system to ensure structured data management, integrity, and efficient querying.

The system is built around a MySQL database where all core entities such as users, pets, cases, matches, and notifications are stored in normalized tables. Relationships between entities are maintained using primary and foreign keys to ensure data consistency.

Key backend functionalities include case registration, pet profile storage, automated match suggestion, and case tracking. The matching system compares attributes such as breed, color, and location to identify potential matches between lost and found reports.

Database constraints such as CHECK constraints and NOT NULL conditions are used to maintain valid data entries. Triggers and procedural logic are implemented to automate actions such as match detection and status updates.

Additionally, location data is stored using latitude and longitude values to support proximity-based queries. Analytical queries are used to generate insights such as recovery rates and case trends.

Overall, the backend ensures reliable data handling, efficient retrieval, and automated processing to support the system's core functionalities.

Contribution of ID: 23301323, Name: Rafid Rohan

- **Designed core database schema (tables, relationships, keys)**
 - **Implemented pet profile and user-related tables**
 - **Developed analytical queries for dashboard insights**
-

Contribution of ID: 23301569, Name: Md. Rahatul Islam Rahat

- **Implemented case registration logic (lost & found cases)**
 - **Developed CRUD operations for case management**
 - **Handled data validation and integrity constraints**
-

Contribution of ID: 23301325, Name: Sk. Md. Rashid Assef Shibly

- **Developed match suggestion logic between cases**
- **Implemented notification handling system**
- **Managed case status updates and relationship linking**

Source Code Repository

<https://github.com/RashidAssef/370-Project>

Conclusion

The Lost & Found Pet Tracking & Recovery Platform successfully demonstrates how a structured database system can solve real-world problems related to lost and found pets. By centralizing scattered information into a unified system, the platform improves efficiency in reporting, tracking, and matching cases.

The system integrates core functionalities such as case registration, pet profile management, automated match suggestion, and status tracking, all supported by a well-designed relational database. Features like location-based querying and case relationship linking further enhance the accuracy and effectiveness of the recovery process.

Throughout the development of this project, key database concepts such as normalization, relational integrity, constraints, and query optimization were applied to ensure reliable and efficient data management.

Overall, the project highlights the importance of database systems in building scalable and practical applications. With further enhancements such as real-time updates and advanced image-based matching, the platform can be extended into a fully deployable solution for real-world use.

References